This contradicts the fact that a & b are co-prime. This contradiction occured due to our wrong assumption.
So,
$\sqrt{3}$ is irrational,
Hence, proved.

Ans. 14. $p(x) = 2x^2 - 3x + 7$

$\alpha + \beta = \frac{-b}{a} = -\frac{(-3)}{2} = \frac{3}{2}$

$\alpha\beta = \frac{c}{a} = \frac{7}{2}$

$\frac{1}{2\alpha - 3} + \frac{1}{2\beta - 3}$

$\frac{(2\beta-3)+(2\alpha-3)}{(2\alpha-3)(2\beta-3)} = \frac{2\beta - 3 + 2\alpha - 3}{4\alpha\beta - 6\alpha - 6\beta + 9}$

$= \frac{2\alpha + 2\beta - 6}{4\alpha\beta - 6\alpha - 6\beta + 9}$ $\cancel{= 2(\alpha+\beta) - 6}$

$= \frac{2(\alpha+\beta) - 6}{14 - 6\left(\frac{3}{2}\right) + 9} = \frac{3 - 6}{14 - 9 + 9} = \boxed{\frac{-3}{14}}$

Ans. 15. For it to have infinite solutions,

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$3x - (a+1)y = 2b - 1$

$5x + (1-2a)y = 3b$

$a_1 = 3,\ b_1 = (-a-1),\ c_1 = -2b+1$

$a_2 = 5,\ b_2 = (1-2a),\ c_2 = -3b$

$\frac{3}{5} = \frac{-a-1}{1-2a}$

$3 - 6a = -5a - 5$

$\boxed{8 = a}$

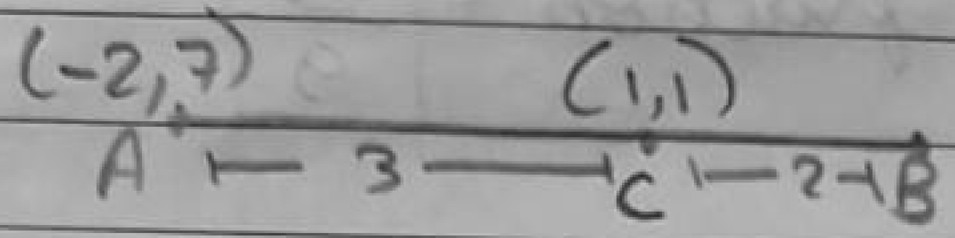
$$\frac{\sin^2\theta}{\sin^2\theta-\cos^2\theta} + \frac{\sin^2\theta\cos^2\theta}{\sin^2\theta(\sin^2\theta-\cos^2\theta)}$$

$$\frac{\sin^2\theta}{\sin^2\theta-\cos^2\theta} + \frac{\cos^2\theta}{\sin^2\theta-\cos^2\theta}$$

$$\frac{\sin^2\theta+\cos^2\theta}{\sin^2\theta-\cos^2\theta} = RHS$$

Hence, proved.

Ans.18. Given,



$C(1,1), A(-2,7)$

$m_1 = 3, m_2 = 2,$

Using section formula,

$$x = \frac{m_1x_2+m_2x_1}{m_1+m_2}$$

$$1 = \frac{3(x_2)+(2)(-2)}{5}$$

$$5 = 3x_2-4$$

$$3x_2 = 9$$

$$\boxed{x_2 = 3}$$

$$y = \frac{m_1y_2+m_2y_1}{m_1+m_2}$$

$$1 = \frac{(3)(y_2)+(2)(7)}{5}$$

$$5 = 3y_2+14$$

$$3y_2 = -9$$

$$\boxed{y_2 = -3}$$

Coordinates of B = $\boxed{(3,-3)}$

Ans.19. let fraction be $\frac{x}{y}$

Given,

$$\frac{2x}{y+1} = 1 \Rightarrow 2x = y+1 \quad \text{——} \ (1)$$

$$\frac{x+4}{2y} = \frac{1}{2} \Rightarrow x+4 = y \quad \text{— (ii)}$$

solving both eq.

$2x - y - 1 = 0$

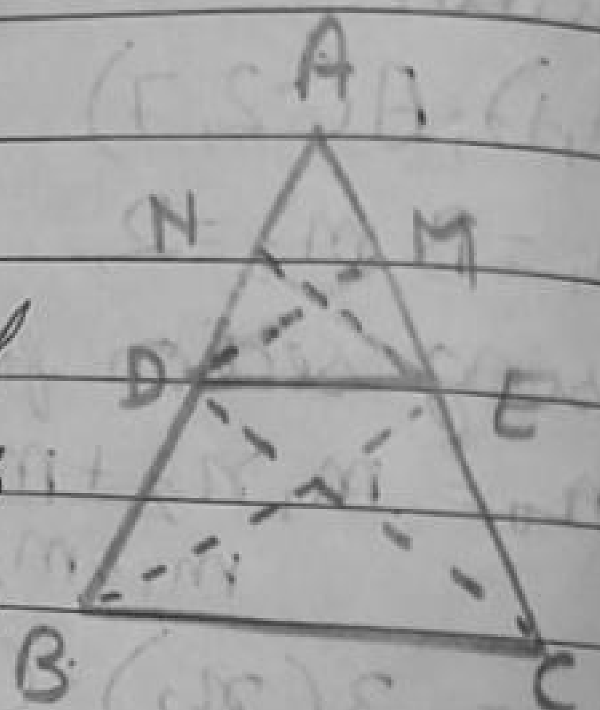
$-x + y - 4 = 0$

$\boxed{x = 5}$

$\boxed{y = 9}$

fraction = $\boxed{\frac{5}{9}}$

Ans. 20. Thales theorem - If a line is drawn parallel to one side of triangle and intersects other two sides at distinct points, then other two sides are divided in same ratio.



Given,

ABC is a tringle

DE || BC,

To prove,

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Construction

Join BE and CD,

draw DM ⊥ AC and EN ⊥ AB,

Proof,

$$\frac{ar(\triangle ADE)}{ar(\triangle BDE)} = \frac{\cancel{1/2} \times AD \times \cancel{EN}}{\cancel{1/2} \times BD \times \cancel{EN}}$$

$$\frac{ar(\triangle ADE)}{ar(\triangle BDE)} = \frac{AD}{BD} \quad \text{— (i)}$$

$$\frac{ar(\triangle ADE)}{ar(\triangle CDE)} = \frac{\cancel{1/2} \times AE \times \cancel{DM}}{\cancel{1/2} \times EC \times \cancel{DM}}$$

$$\frac{ar(\triangle ADE)}{ar(\triangle CDE)} = \frac{AE}{EC} \quad \text{—— (II)}$$

$ar(\triangle BDE) = ar(\triangle CDE)$ —— (III)

⟶ [$\triangle$'s on same base, between same parallel lines]

from eq (I), (II) and (III),

$$\boxed{\frac{AD}{DB} = \frac{AE}{EC}}$$

Hence, proved.

Ans. 21

$$\frac{\sin 30^\circ + 2\cos^2(45) + \tan^2(60)}{\frac{1}{2}\cot 45 + \cos^2(30) + \tan^2(45)}$$

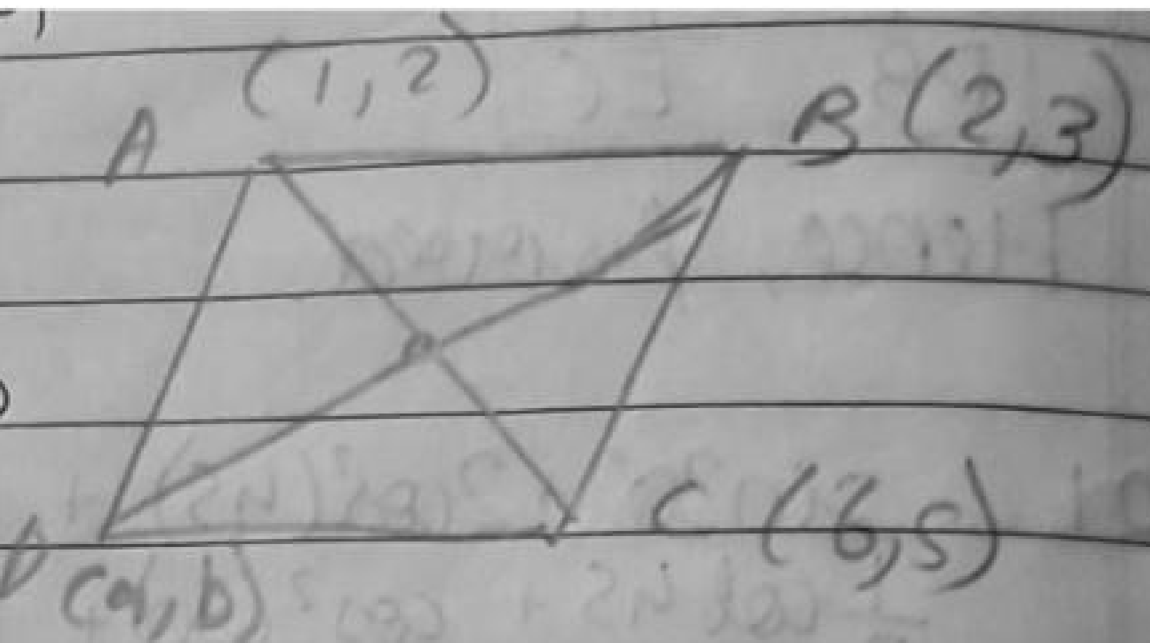
$$= \frac{\frac{1}{2} + 2 \cdot \left(\frac{\sqrt{2}}{2}\right)^2 + \tan^2(60)}{\frac{1}{2}\cot 45 + \cos^2(30 + \tan^2(45)}$$

$$= \frac{\frac{1}{2} + \cancel{2} 1 + (\sqrt{3})^2}{\frac{1}{2}\cot 45^\circ + \cos^2(30) + \tan^2(45)}$$

$$= \frac{9/2}{\frac{1}{2} + \frac{3}{4} + \tan^2(45)}$$

$$= \boxed{2}$$

$-10x - 6y = -4$



Ans. 22. Join AC and BD,

O is midpoint of AC & BD

so,

midpoint of AC

$\hookrightarrow \left(\frac{6+1}{2}, \frac{5+2}{2}\right) = \left(\frac{7}{2}, \frac{7}{2}\right)$

midpoint of BD

$\hookrightarrow \left(\frac{2+a}{2}, \frac{3+b}{2}\right)$

~~so~~ As both points are same,

$\frac{2+a}{\cancel{2}} = \frac{7}{\cancel{2}}$

$\boxed{a = 5}$

$\frac{3+b}{\cancel{2}} = \frac{7}{\cancel{2}}$

$\boxed{b = 4}$

coordinates of D = (5, 4)

Ans. 23. A will complete 1 round $= \frac{1980}{330} = 6$ min

B " " " $= \frac{1980}{198} = 10$ min